

Discrete Structures

CSCI 246

Instructor Info

Dr. Michael Wojnowicz (Mike)

Office Hrs: MW 3-4, Fr 1-2

Gianforte Hall 360B

https://mikewojnowicz.github.io/

michael.wojnowicz@montana.edu

Course Info

Prereq: Calculus, although this is not used.

Class meetings: Mon, Wed, Fri 2:10-3p

Reid 201

TA Info

Madhav Cherupilil

Thurs 3-6pm; Fri AM by appt.

Tutoring Center (GH 118)

m.cherupililsajeev@montana.edu

Tutor Info

Holden Lee

holdenhlee123@gmail.com

Mon-Fri 3:30-5:30. By appointment. Book here. Sessions are free!

SmartyCats Center (2nd floor Renne).

Course description

Mathematical concepts used in computer science with an emphasis on mathematical reasoning and proof techniques. This course covers mathematical thinking (elementary logic, theorems, proof techniques), discrete collections (such as sets and lists), functions and relations, counting (including permutations and combinations), and miscellaneous topics (recursions, Big-O notation, graphs). Writing and understanding proofs will be an important part of this course.

Readings

Primary textbook

Scheinerman, E. R. (2013). *Mathematics: a discrete introduction.* Third edition.

Supplemental readings

For certain topics, I will provide handouts, drawn from:

- Susanne Epp (2020). *Discrete Mathematics with Applications*.
- Lovász, Pelikán, and Vesztergombi (2003). *Discrete Mathematics: Elementary and Beyond.*
- Joel David Hamkins (2020). *Proof and the Art of Mathematics.*

Learning Outcomes

- To learn to reason mathematically and communicate ideas in a clear and concise manner;
- To use mathematical proofs.
- To understand several central concepts from discrete mathematics that are used throughout computer science;

Learning Philosophy

This course will use an “inquiry-based learning" approach. This means that classes will not be lecture-oriented. Instead, to quote Dr. Dana Ernst,

> *You will be expected to work actively to construct your own understanding of the topics at hand with the readily available help of me and your classmates. Many of the concepts you learn and problems you work on will be new to you and ask you to stretch your thinking. You will experience frustration and failure before you experience understanding. This is part of the normal learning process. If you are doing things well, you should be confused at different points in the semester.*

Active learning has been shown to increase student performance in STEM courses (e.g. see [1, 2]).

Class activity

A typical class meeting will be structured as follows:

Mon/Wed

- Review exercises: 15 mins
- Mini-lecture: 20 mins
- Group exercises: 15 mins

Fri

- Weekly quiz: 20 mins
- Review exercises: 15 mins
- Mini-lecture: 15 mins

Daily readings

Before each class meeting, you will be assigned a reading from the textbook. The reading serves preparation for the group exercises.

Group exercises

The group exercises are for collaborative problem-solving. Each class meeting, students will be split randomly into groups of three to work on problems using the whiteboards (or windows). Mathematics is not a spectator sport!

Weekly quiz

The Friday weekly quizzes will cover group exercises and the readings from a batch of three class meetings (that week’s Mon., and the previous Wed. and Fri.)

FAQs

What are discrete structures?

A discrete structure refers to a mathematical system that is composed of distinct, separate elements, as opposed to continuous structures where elements can vary smoothly. Think of a digital clock vs. an analog clock (where second hand loops around continuously without stopping). Examples of discrete structures include sets with finitely many elements (e.g. the integers 1 to 10), lists, graphs, and logical statements.

What is discrete mathematics?

Discrete mathematics is the study of discrete structures and mathematical operations that can be performed upon them.

Why study discrete mathematics?

It is the mathematics underlying almost all of computer science. Here are a few examples: (1) designing high-speed networks and message routing paths, (2) finding good algorithms for sorting, (3) performing web searches, (4) analyzing algorithms for correctness and efficiency, (5) formalizing security requirements, and (6) designing cryptographic protocols.

Where is discrete mathematics used in the MSU computer science curriculum?

Discrete mathematics is used throughout the curriculum. Proofs in particular play a critical role in CSCI 338 (Computer Science Theory) and CSCI 432 (Advanced Algorithm Topics).

Grading

- Participation: 10%
- Weekly Quizzes: 30%
- Midterm: 20%
- Final: 25%
- Project: 15%

Grades will be assigned as follows:
A: 93-100, A-: 90-93, B+: 87-90, B: 83-87, B-: 80-83, C+: 77-80, C: 73-77, C-: 70-73, D+: 67-70, D: 63-67, D-: 60-63, F: 0-60.

Mini-lectures. I plan to post slides before class so that you can use them as reference during the mini-lectures

Project. The purpose of the project is for students to focus on some aspect of discrete mathematics of particular interest. Students will work in small groups, choose some aspect of discrete mathematics (that was not covered in class), and develop an 8-10 minute presentation to give to the class. For example, a group might give an overview to public key cryptography, or explore how graph theory underlies web search. There will also be a written component. More information will be given as the time approaches.

Scholarly Responsibilities. The class has no graded homework. However, to be successful in the course, you need to do the following:

- Before class: *Read the assigned section.* Be sure to read actively , with a pencil and paper in hand. Work out the examples within the section on your own, using the textbook to check your work. You do NOT need to do the textbook exercises (although you may).
- During class: *Attend class.* This allows you to participate in the mini-lecture discussions and group exercises.
- After class: *Finish working through the group exercises.* Solutions to group exercises are posted within an hour after class. After class, you should
  1. Make sure you know how to do all the group exercises.
  2. Make sure the form of your solution matches the provided solutions (e.g. in terms of amount of justification).

Makeup policy. Quizzes and participation have no makeups. However, I drop 4 participation days and 2 weekly quizzes with no consequence. Missed material beyond that (with a valid explanation and documentation) can be replaced with the final exam grade.

Diversity and Inclusivity Statement. I consider this classroom to be a place where you will be treated with respect, and I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability - and other visible and non-visible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. Your suggestions about how to improve the value of diversity in this course are encouraged and appreciated. Please let me know ways to improve the effectiveness of the course for you personally or for other students or groups.

Accommodations for Students with Disabilities. If you are a student with a disability and wish to use your approved accommodations for this course, contact me during my office hours to discuss. Please have your Accommodation Notification available for verification of accommodations. Accommodations are approved through the Office of Disability Services located in 137 Romney Hall. www.montana.edu/disabilityservices.

Student Conduct. You are expected to abide by MSU's Code of Student Conduct.

Class Schedule

MODULE 0: Course Overview

Wed	Aug 26	Course Overview	

MODULE 1: Fundamentals

Fri	Aug 28	Definition	Sec 3
Mon	Aug 31	Theorem	Sec 4
Wed	Sept 2	Proof	Sec 5
Fri	Sept 4	Counterexample	Sec 6
Mon	Sept 7	Holiday (Labor Day)	
Wed	Sept 9	Boolean Algebra	Sec 7

MODULE 2: Discrete Collections

Fri	Sept 11	Lists	Sec 8
Mon	Sept 14	Factorial	Sec 9
Wed	Sept 16	Sets	Sec 10
Fri	Sept 18	Quantifiers	Sec 11
Mon	Sept 21	Operations on Sets	Sec 12

MODULE 3: Counting and relations

Wed	Sept 23	Relations	Sec 14
Fri	Sept 25	Equivalence Relations	Sec 15
Mon	Sept 28	Partitions	Sec 16
Wed	Sept 30	Binomial Coefficients (Combinations)	Sec 17

MIDTERM

Fri	Oct 2	Midterm Review Part 1	
Mon	Oct 5	Quiz Retake (Optional); Midterm Review Part 2	
Wed	Oct 7	MIDTERM EXAM	

MODULE 4: More proof

Fri	Oct 9	Contrapositive and contradiction	Sec 20
Mon	Oct 12	Smallest counterexample	Sec 21
Wed	Oct 14	Induction	Sec 22
Fri	Oct 16	Recurrence	Sec 23

Mon	Oct 19	Multiple Proofs	Hamkins Ch. 2
MODULE 5: Functions and Big O notation			
Wed	Oct 21	Functions	Sec 24
Fri	Oct 23	Real-valued functions	Epp Sec 11.1
Mon	Oct 26	Catchup Day	None
Wed	Oct 28	Big O Notation	Epp Sec 11.2
Fri	Oct 30	Application: Analysis of Algorithm Efficiency	Epp Sec 11.3
MODULE 6: Discrete probability			
Mon	Nov 2	Intro to Probability (Part 1)	Sec 30
Wed	Nov 4	Intro to Probability (Part 2)	Sec 31
Fri	Nov 6	Conditional Probability /Independence	Sec 32
Mon	Nov 9	Random variables	Sec 33
Wed	Nov 11	Holiday (Veterans Day)	
Fri	Nov 13	Expectation	Sec 34
MODULE 7: Graph theory			
Mon	Nov 16	Fundamentals of Graph Theory	Sec 47
Wed	Nov 18	Subgraphs	Sec 48
Fri	Nov 20	Connection	Sec 49
	Nov 23-27	Fall Break	N/A
Mon	Nov 30	Eulerian Graphs	Sec 51
MODULE 8: Group Presentations			
Wed	Dec 2	Prepare Group Presentations	
Fri	Dec 4	Group Presentations	
Mon	Dec 7	Group Presentations	
Wed	Dec 9	Group Presentations	
FINAL			
Fri	Dec 11	Final Preparation	
Wed	Dec 16	Final (2:00 - 3:50pm)	

Note: The class schedule is tentative; it is subject to change as the course progresses.

External Resources

For a list of external resources on discrete mathematics, click here. This list will be updated throughout the semester.

References

[1] Louis Deslauriers, Ellen Schelew, and Carl Wieman. Improved learning in a large-enrollment physics class. *science*, 332(6031):862–864, 2011.

[2] Scott Freeman, Sarah L Eddy, Miles McDonough, Michelle K Smith, Nnadozie Okoroafor, Hannah Jordt, and Mary Pat Wenderoth. Active learning increases student performance in science, engineering, and mathematics. *Proceedings of the national academy of sciences*, 111(23):8410–8415, 2014.